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ORIGINAL ARTICLE

REMFLOW: RAG-enhanced multi-factor rainfall flooding warning in sponge airports via large language model

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Abstract

The integration of rainfall flooding data, particularly with the development and application of the storm water management model (SWMM), has significantly enhanced the predictive capabilities of flood warning systems. However, existing models often lack relevant environmental context, leading to the incorporation of external knowledge bases. Previous machine learning approaches have primarily focused on structured knowledge, neglecting unstructured data and high-dimensional semantic information. To address these issues, we propose REMFLOW, a retrieval-augmented generation (RAG) framework designed to improve the analysis and application of multi-factor rainfall flooding data. We employ an embedding model to encode time-series flood data, which is then stored in a knowledge base. Simultaneously, a large language model (LLM) is used to encode multi-factor rainfall flooding data and retrieve relevant knowledge. Subsequently, an adaptive context fusion mechanism is applied to integrate the extracted knowledge and generate the final outputs. Experimental results from various rainfall flooding prediction tasks demonstrate that the REMFLOW framework significantly outperforms baseline models in the precise predictive analysis of multi-factor rainfall flooding data, highlighting its effectiveness.

Keywords Sponge airports · Retrieval-augmented generation · Rainfall flooding warning · Large language model

1 Introduction

The application of SWMM[1] and sponge airport design principles in flood early warning systems marks a significant advancement in simulating, analyzing, and collecting rainfall data, providing a strong foundation for more effective flood prediction and management [34]. This progress is largely attributed to using multi-factor stormwater management data, including rainfall records, runoff data, and various environmental parameters specific to sponge airports. These data types are crucial for flood prediction tasks, reflecting a comprehensive approach that leverages multiple environmental data points for decision-making and strategy development, rather than relying on a single data source.

Machine learning methods have become central to processing multi-factor data, effectively mapping heterogeneous inputs to predictive outputs. However, unlike experienced hydrologists who possess deep environmental insights, neural networks trained from scratch often lack an understanding of hydrological and environmental concepts [39]. Without integrating external knowledge, these networks typically struggle to identify key environmental factors or anomalies in the data, which are crucial for accurate predictions. Recent studies have addressed this by incorporating knowledge bases into analyses, providing additional environmental context [20]. These knowledge bases offer supplementary layers of information, enhancing the model's ability to generate meaningful representations and interpretable evidence.

Previous approaches to integrating external environmental knowledge with stormwater data analysis, particularly in the context of SWMM and sponge airport designs, have primarily focused on extracting hierarchical and structured knowledge from specific knowledge bases. These methods mainly incorporate environmental concepts into structured frameworks, contributing minimally to predictive tasks

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[1] https://www.epa.gov/water-research/storm-water-management-model-swmm.

[9]. Furthermore, they often rely exclusively on structured data, such as rainfall amounts, drainage capacity, and environmental conditions, while neglecting unstructured data. Although more challenging to extract, unstructured data, such as known records and real-time sensor readings, are more prevalent and practical. Given the strong capabilities of large language models (LLMs) in various predictive tasks and their potential use as extensive environmental knowledge bases, it is feasible to integrate complex environmental knowledge from these sources with multi-factor stormwater data. However, this approach must address the challenge of hallucinations in LLMs, where incorrect or fabricated information may arise.

To overcome these limitations, we propose REMFLOW, a method that utilizes LLM within a RAG approach for flood early warning in sponge airports. The methodology and input examples we constructed are presented in Fig. 1. We employ the RAG process to link unstructured flood-related data as context with a multi-factor flood data knowledge base, leveraging the semantic reasoning capabilities of LLMs for data analysis and prediction across different sponge airports and various regions within the same airport. This approach aims to overcome challenges related to the complexity of data elements, the lack of comprehensive datasets, and the inaccuracies in data analysis and intelligence, as well as to mitigate the hallucination problem that LLMs may exhibit in vertical domains. The main contributions are summarized as:

- We designed a multi-factor data structure for flood early warning in sponge airports based on the SWMM model and constructed a high-quality dataset that covers different sponge airports and various regions within the same

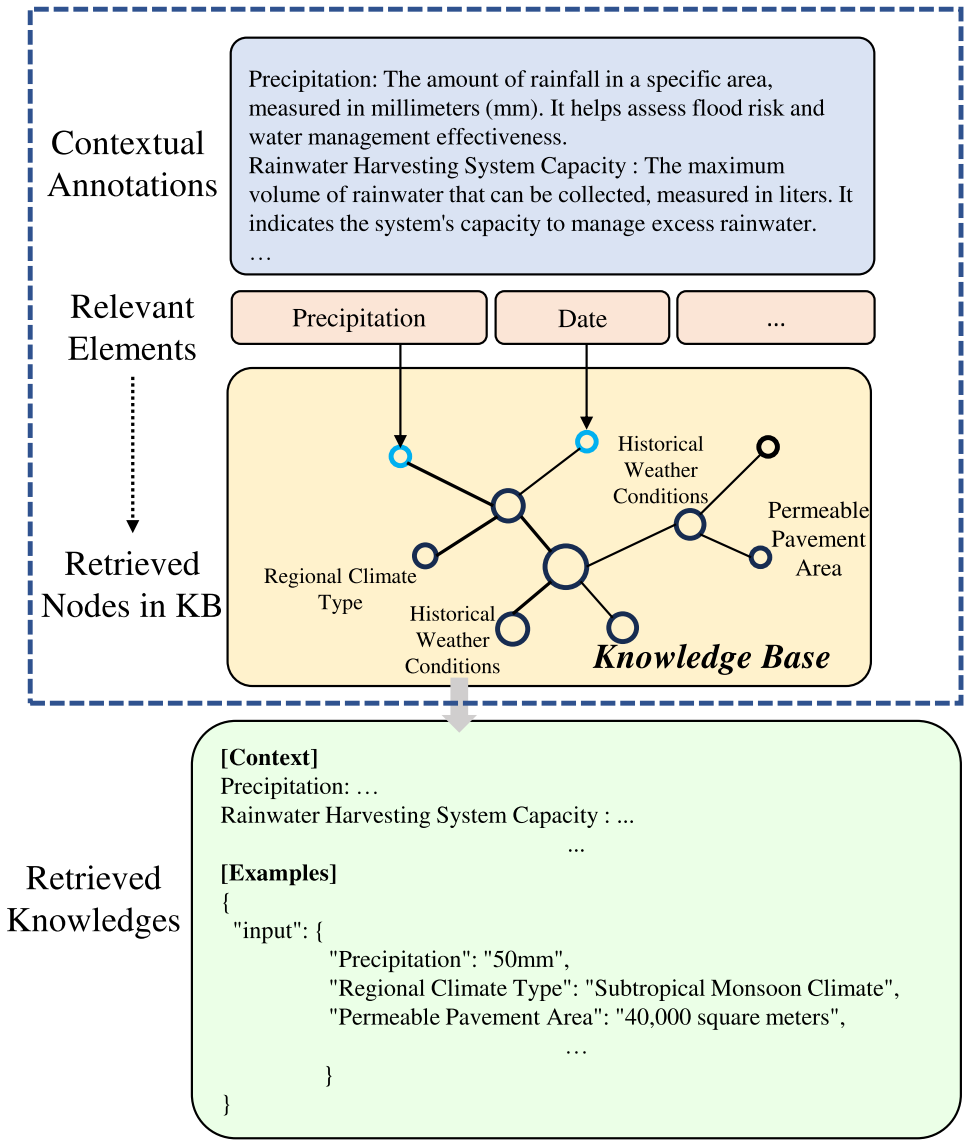


Fig. 1 Construction of input content for the large language model

airport. This dataset is pioneering and ensures practical utility and relevance.

- We used the dataset as a knowledge base to provide prior knowledge of the LLM and enhanced the input context with large-scale LLMs, enabling the model to produce more effective outputs. This expands the application of LLMs in vertical domains, particularly in flood early warning.
- We propose REMFLOW, a RAG-based LLM approach for flood data analysis and early warning. This method improves the accuracy of predictions and enhances the intelligence of sponge airport warnings. Comparative experiments with traditional machine learning methods demonstrate the practical significance of REMFLOW.
- In the evaluation of flood warning performance, we conducted a comprehensive assessment of REMFLOW by combining prediction accuracy and semantic similarity. The evaluation includes comparisons with traditional machine learning methods and analyzes different dataset scales, providing valuable insights for future research.

2 REMFLOW method

2.1 Model overview

In Fig. 2, we outline a retrieval-augmented generation (RAG) framework based on large language model (REMFLOW) for intelligent warning systems in sponge airports. We utilize GPT-4o for data augmentation to transform specialized knowledge into formats that the models can better understand [13]. By combining outputs from GPT-4o with manual verification, we construct datasets tailored to different regional sponge airport conditions and rainfall scenarios within the same airport. These datasets support intelligent early warnings under low, moderate, and high-risk conditions. Then, we utilize the vectorization model to transform text data into vectors. These vectors are stored in a knowledge base constructed with the vector database. We calculate vector similarity using Euclidean distance to retrieve the closest k data points for the queries. Finally, we input the retrieved information and context into open-source large models like Qwen2 [38] for the efficient execution of rainwater flood prediction and recommendation tasks. This method forms a comprehensive warning system, which we compare against traditional machine learning approaches to demonstrate its superiority.

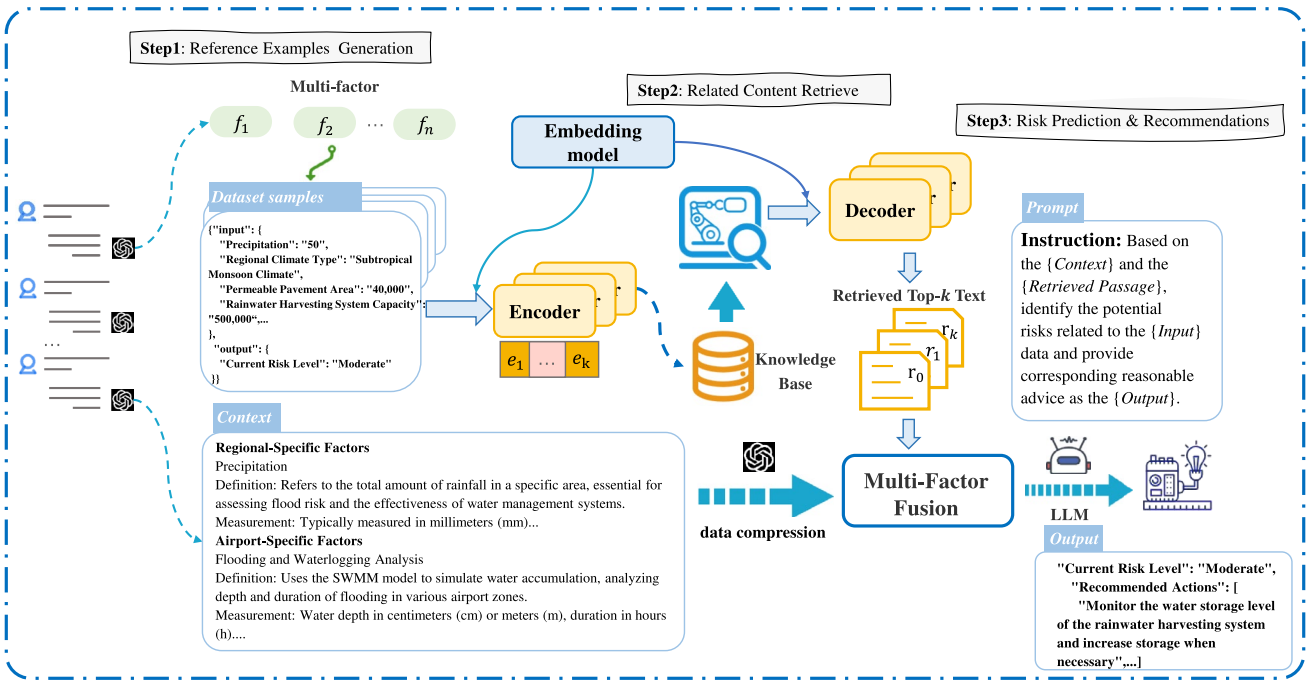


Fig. 2 Overview of REMFLOW. We utilize language model tools (GPT-4o) to construct key elements f_1 to f_n for sponge airports based on the SWMM model and relevant knowledge. We then gather information from these key elements to create early warning data. An embedding model maps text content into k-dimensional vectors for efficient storage and retrieval. To enhance information integration, we compress the explanations of multiple key elements as context. Finally, we merge the retrieved data and context, enabling the large language model to conduct rainwater flood predictions and recommendations

2.2 Datesets construction

Definition of multi-factor As shown in Fig. 3, the SWMM-based flood simulation structure illustrates typical hydraulic components and drainage systems. Key elements such as subcatchments, junctions, dividers, and storage units simulate flood flow paths and drainage responses, supporting multi-zone storm simulations in sponge airport environments. By leveraging the interaction between human experts and SWMM model, we developed a multi-factor framework for professional rainfall and flood warning. This framework encompasses 9 essential *regional-specific factors* for sponge airports in different regions, such as Precipitation, Date, Regional Climate Type, and Historical Weather Conditions. For distinct zones within the same airport, we established 5 *airport-specific factors*, including Storm Event Simulation and Drainage System Process. We further standardized the definitions of these factors through data augmentation, ensuring their suitability as contextual inputs for the model, as outlined in Table 1. This multi-factor approach provides a robust foundation for constructing high-quality datasets.

Datesets generation Based on the already generated regional-specific factors and airport-specific factors, we utilize GPT-4o, which possesses extensive training data, supports longer contexts, and demonstrates powerful generative capabilities, to assist in data generation. We input collected information and definitions of key factors to extract essential insights, enabling the construction of data in a specified JSON format for practical application and processing. For the regional-specific factors dataset, characterized by fixed values for each key element, we generated 1000 data entries representing various climate regions relevant to flood and rainfall warnings at sponge airports. On the other hand, the airport-specific factors dataset employs attribute values for each factor, resulting in 200 entries for each of the three distinct zones within a sponge airport: *Airfield*, *Terminal Area*, and *Ground Transport Area*, which yields a total of 600 entries. This methodology effectively captures variability and uncertainty, enhancing the realism of the datasets. For the output, the specified model classifies risk levels into three categories: {Low}, {Moderate} and {High}, while also providing recommendations for appropriate response measures. The design of the prompts used for data generation is detailed in Appendix 1.1. Given that the rainfall and flood warning system is primarily designed to address severe weather conditions, our collected and generated data distribution contains a significant proportion of entries categorized with a High-risk level. In total, we produced 1600 validated data entries, providing a robust foundation for subsequent experimental designs, with sample data formats presented in Appendix 2.1.

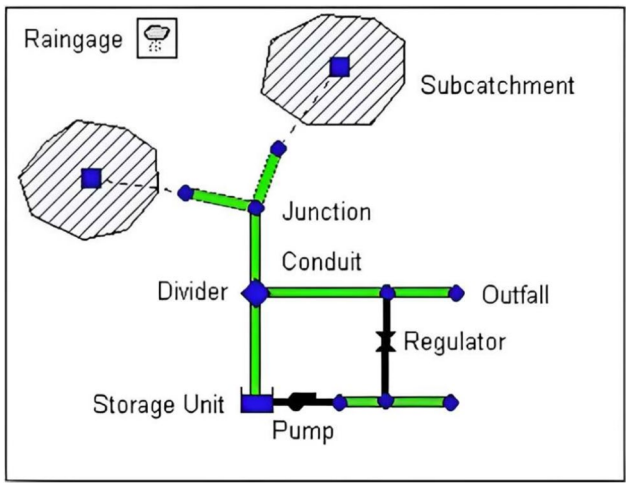


Fig. 3 SWMM model structure for flood simulation

2.3 RAG-enhanced large language model

In this study, we apply the vectorization model to extract embeddings from structured data in flood warning systems, facilitating efficient retrieval and feature extraction within an enhanced retrieval-augmented generation (RAG) framework. The input data $\boldsymbol{f}$ includes multiple key factors, such as regional, weather, and temporal information. For both region data $\boldsymbol{f}_{\text{Area}}$ and date data $\boldsymbol{f}_{\text{Date}}$, along with other continuous factors, we utilize a Multilayer Perceptron (MLP) for encoding. Additionally, a sinusoidal transformation inspired by the Transformer positional embedding mechanism is applied to generate a unified multi-factor embedding $\boldsymbol{e}_{\text{Factors}}$. This approach not only captures the intricate relationships between different regions but also effectively represents the temporal characteristics of dates and other continuous variables.

$$\boldsymbol{e}_{\text{Factors}} = \text{MLP}(\boldsymbol{f}_{\text{Area}}, \boldsymbol{f}_{\text{Date}}, \ldots) \tag{1}$$

Data for analysis $\boldsymbol{Q}_{\text{Text}}$ is encoded using a large scale embedding model (LEM), generating the text embedding $\boldsymbol{e}_{\text{Text}}$, which is well-suited to capture rich contextual information present in textual records.

$$\boldsymbol{e}_{\text{Text}} = \text{LEM}(\boldsymbol{Q}_{\text{Text}}) \tag{2}$$

After embedding the input data using the vectorization model, we apply a RAG-based framework to enhance flood warning predictions for sponge airports. The embedded query data $\boldsymbol{Q}$, including region, weather, and sponge airport characteristics, is used to retrieve relevant known data from the knowledge base. This retrieval process matches the current query with past records that share similar multi-factor

Table 1 List of the factors for sponge airport flood warning

Factor name	Factor description
Regional-specific factors	
(1) Precipitation	**Definition**: Refers to the total amount of rainfall in a specific area, essential for assessing flood risk and water management effectiveness
	Measurement: Typically measured in millimeters (mm)
(2) Date	**Definition**: The specific date when the weather data is recorded. This helps in correlating weather patterns and assessing potential risks
	Measurement: Expressed as a calendar date (e.g., YYYY-MM-DD)
(3) Regional climate type	**Definition**: The general classification of the climate in the region, such as "Subtropical Monsoon Climate." It helps in predicting regional weather patterns and responses to weather events
	Measurement: Described using standard climate classifications (e.g., subtropical, temperate)
(4) Historical weather conditions	**Definition**: Information about past weather patterns, including rainfall frequency and intensity. Used to identify trends and predict future events
	Measurement: Quantified using historical data on rainfall events, often summarized by averages or extremes
(5) Permeable pavement area	**Definition**: The surface area covered by permeable materials that allow water to pass through, reducing runoff and aiding infiltration in flood management
	Measurement: Measured in square meters (m^2)
(6) Rainwater harvesting system capacity	**Definition**: The maximum volume of rainwater that can be collected and stored by the system. It indicates the system's ability to manage excess rainfall
	Measurement: Measured in liters (L)
(7) Vegetation coverage rate	**Definition**: The percentage of the area covered by vegetation. Vegetation helps absorb water, reducing runoff and mitigating flood risks
	Measurement: Expressed as a percentage (%)
(8) Retention pond capacity	**Definition**: The maximum volume of water that can be held by retention ponds. These ponds are used to temporarily store stormwater runoff, reducing the risk of flooding
	Measurement: Measured in liters (L)
(9) Groundwater recharge rate	**Definition**: The volume of water added to the groundwater system on a monthly basis. This is critical for maintaining groundwater levels, especially during dry periods
	Measurement: Measured in cubic meters per month (m^3/month)
Airport-specific factors	
(1) Storm event simulation	**Definition**: Simulates storm events using the SWMM model to analyze rainfall distribution and water accumulation in different airport areas
	Measurement: Based on historical rainfall data and current forecasts, typically expressed in millimeters per hour (mm/h)
(2) Flooding and waterlogging analysis	**Definition**: Uses the SWMM model to simulate water accumulation, analyzing depth and duration of flooding in various airport zones
	Measurement: Water depth in centimeters (cm) or meters (m), duration in hours (h)
(3) Drainage system process	**Definition**: Analyzes and monitors the efficiency of the airport's drainage system during storms, including pipe flow rate and drainage speed
	Measurement: Pipe flow rate in liters per second (L/s) or cubic meters per second (m^3/s), drainage speed in meters per second (m/s)
(4) Real-time hydrological monitoring	**Definition**: Monitors real-time hydrological data such as rainfall, water levels, and water quality within the airport, providing dynamic inputs for the SWMM model
	Measurement: Rainfall in millimeters (mm), water level in meters (m), water quality parameters like pH and dissolved oxygen in specific values
(5) Waterlogging point warning indicators	**Definition**: Identifies key waterlogging points within the airport and sets corresponding warning thresholds
	Measurement: Determined by water depth, duration, and drainage capacity, may include geographic location and depth thresholds for waterlogging points

structures. For each feature in the query data $\boldsymbol{Q}_{\text{Factors}}$, we compute its deviation from the corresponding known data in the knowledge base by calculating the z-score s_i for each feature $\boldsymbol{Q}_i$:

$$s_i = \frac{\boldsymbol{Q}_i - \text{mean}(\boldsymbol{K}_i)}{\text{std}(\boldsymbol{K}_i)} \tag{3}$$

Here, $\boldsymbol{K}_i$ represents the known data of the corresponding feature. The z-score highlights deviations from known patterns, helping us identify critical features for retrieval. Instead of cosine similarity, we use Euclidean distance to measure the closeness between the query's embedding $\boldsymbol{e}_{\text{Factors}}$ and known data embeddings $\boldsymbol{h}_{\text{KB}}$ from the knowledge base. The Euclidean distance d_q^n is calculated as:

$$d_q^n = \|\boldsymbol{e}_{\text{Factors}} - \boldsymbol{h}_{\text{KB}}\| \tag{4}$$

The known records with the smallest Euclidean distance are considered the most relevant. These records provide critical context, such as similar rainfall patterns or sponge airport infrastructure performance during previous flood events.

2.4 Multi-factor fusion module

We extend the RAG framework to integrate key textual data representations, denoted as $\boldsymbol{h}_{f_1}, \boldsymbol{h}_{f_2}, \ldots, \boldsymbol{h}_{f_n}$. These represent various factors such as weather, contextual information, and other elements relevant to flood prediction.

First, we concatenate these representations and use an MLP to map them into a unified embedding. The contextual information is compressed using LLM, ensuring efficient feature extraction for accurate predictions.

$$\boldsymbol{h'} = \text{MLP}(\text{Concat}(\boldsymbol{h}_{f_1}, \boldsymbol{h}_{f_2}, \ldots, \boldsymbol{h}_{f_n})) \tag{5}$$

We propose a self-attention-based fusion network to capture dependencies between these factors. Self-attention is applied to each factor's representation to enhance the integration of information.

$$\tilde{h} = \text{MHSA}(h' + h_{\text{Context}}) \tag{6}$$

In this network, MHSA refers to multi-head self-attention, and h_{Context} represents the contextual embedding derived from the compressed data. Residual connections and BatchNorm are applied to stabilize the learning process.

The output of the self-attention layer is processed through an attention pooling layer to extract the most critical information for flood warning.

$$\boldsymbol{Z} = \text{AttnPool}(\text{MLP}(\tilde{h})) \tag{7}$$

As shown in Fig. 4, the final fused representation $\boldsymbol{Z}$ is used to generate flood warning predictions for sponge airports.

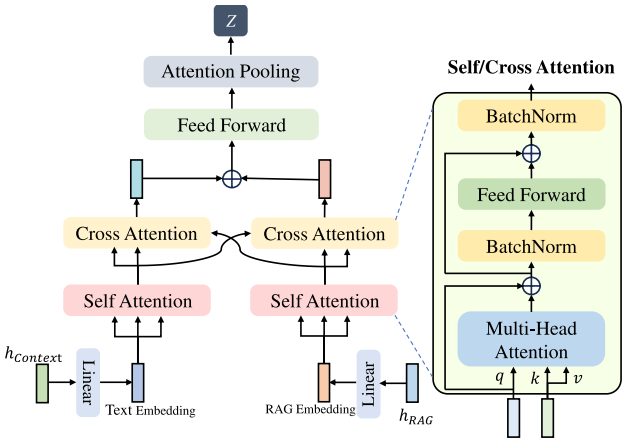


Fig. 4 Flood warning multi-factor fusion module. It integrates multiple textual factors into a fused representation for intelligent flood prediction

3 Experiment setting

3.1 RAG pipeline

Backbone models Based on bilingual large language models in both Chinese and English, and supporting long context, as well as easy migration and deployment, we select foundation models from Qwen2.5-Instruct-7B/14B,[2] InternLM2-7B-Chat,[3] Llama-2-13B-chat,[4] Llama3.1-8B-Instruct,[5] Deepseek-llm-7B-chat[6] and Mistral-7B-Instruct-v0.2[7] to build REMFLOW. These foundation models all support loading within the vllm [18] framework. These models have the ability to support long texts and strong instruction-following capabilities, can be deployed offline, and can enhance the protection of sensitive data in vertical fields. The descriptions of the various models are presented in Table 2, where 1k represents 1, 024 tokens.

Implementation details For Machine Learning Models, we use the AutoGluon [8] framework, which allows for easy invocation of various models and automatic hyperparameter tuning, ultimately yielding the optimal predictive results. For the parameter settings of large language models, to ensure configuration consistency, we used the default settings from the **openai** library, including parameters such as **temperature** (to adjust the randomness of model outputs) and **repetition penalty** (to prevent repetitive outputs).[8] We set the parameter k for the number of retrieved samples in RAG to

[2] https://github.com/QwenLM/Qwen2.5.
[3] https://huggingface.co/internlm/internlm2-7b.
[4] https://huggingface.co/meta-llama/Llama-2-13b-chat.
[5] https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct.
[6] https://huggingface.co/deepseek-ai/deepseek-llm-7b-chat.
[7] https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.2.
[8] https://github.com/openai/openai-python.

Table 2 Advantages of various large language models

Model name	Max token length	Language support	Advantages
Qwen2.5-Instruct-7B/14B	128k tokens	Multilingual	Excellent instruction following and versatility in applications
InternLM2-7B-Chat	200k tokens	Multilingual	Optimized for conversational AI with **long context support** and efficient performance
Llama-2-13B-chat	4k tokens	Multilingual	Strong performance across various NLP tasks and open-source flexibility
Llama3.1-8B-Instruct	128k tokens	Multilingual	Improved instruction following with enhanced understanding capabilities
Deepseek-llm-7B-chat	32k tokens	Multilingual	Specialized in domain-specific applications with optimized chat performance
Mistral-7B-Instruct-v0.2	32k tokens	Multilingual	Efficient model with strong instruction following and adaptability

the commonly used value of 5. A recall length that is too long may increase the model's difficulty in understanding, while a recall length that is too short may result in insufficient information [21].

Knowledge base We use FAISS[9] as the vector knowledge database, which, based on the Langchain architecture,[10] allows for efficient vectorization, storage, and retrieval of structured data. BGE-M3[4] is then used as the vectorization model. This model produces 1024-dimensional vectors, surpassing the input length limitations of other pre-trained models like BERT and achieving higher precision than standard embedding models with 768 dimensions.

Dataset The datasets based on regional-specific factors and airport-specific factors are divided at an 8:2 ratio into Known data (Train data), which includes both "Input" and "Output," and Unknown data (Test data), which contains only "Input." The Known data (Train data) is used for indexing/training, while the Unknown data (Test data) is used to test the predictive capabilities of REMFLOW and traditional machine learning methods. This is in line with the conventional method of dividing training sets and test sets in the field of machine learning, and after our data set is divided according to this ratio, both training and testing can be more fully reflected [7].

3.2 Evaluation

Accuracy score We use accuracy (Acc) to evaluate the flood warning tasks at the Sponge Airport. The flood warning system classifies the results into three categories: low, moderate, and high. The accuracy is calculated by comparing the predicted warning levels with the actual levels across all data points, measuring how often the system correctly identifies the correct category out of the three.

F1 score We use the macro F1 score to evaluate the system's prediction performance. We calculate precision (P_i) and recall (R_i) individually for each category i. Precision (P_i) is defined as the ratio of true positives for category i to the sum of true positives and false positives for that category, and recall (R_i) is the ratio of true positives for category i to the sum of true positives and false negatives for that category. The macro F1 score is then computed by calculating the F1 score for each category and taking the average across all categories, using the following formula (N is the number of categories):

$$F_1 = \frac{1}{N}\sum_{i=1}^{N} 2 \times \frac{P_i \cdot R_i}{P_i + R_i}$$

Rouge-L score Rouge-L is used to evaluate the recommendations generated by the large language model (LLM) in flood risk warnings across different levels. It measures the longest common subsequence (LCS) between the generated and reference recommendations, reflecting both precision and recall. Precision is calculated as the ratio of the LCS length to the length of the generated recommendation, while recall is the ratio of the LCS length to the reference recommendation length. The final Rouge-L score is given by:

$$\text{Rouge-L} = \frac{(1+\beta^2) \cdot \text{Precision} \cdot \text{Recall}}{\beta^2 \cdot \text{Precision} + \text{Recall}}$$

4 Main results

Evaluation results In Table 3, we compare two approaches: training traditional machine learning models using numerical features from the dataset, and inputting structured data as known knowledge into large language models via retrieval. The Acc and F1 results show that REMFLOW, based on large language models (especially Qwen2.5), significantly outperforms traditional machine learning methods in prediction accuracy. Additionally, as the model size increases, the performance further improves. For the two datasets, regional-specific factors has more elements but fewer features, where traditional machine learning performs better than on the airport-specific factors dataset, reflecting its focus on relationships between features rather than deeper

[9] https://ai.meta.com/tools/faiss/.

[10] https://github.com/langchain-ai/langchain.

semantic understanding. In contrast, REMFLOW, leveraging large models and RAG, shows minimal performance differences across both datasets, demonstrating superior generalization and semantic comprehension. This highlights the applicability of REMFLOW for various analytical tasks in sponge airport flood risk prediction.

Ablation studies To evaluate the effectiveness of each component in our proposed method, we conducted an ablation study. In Table 4, the ablation study based on Qwen2.5-14B-Instruct and Llama3.1-8B-Instruct, exemplary representatives of Chinese and English Large Language Models, demonstrates the importance of each component in REMFLOW. To evaluate the effectiveness of each component in our proposed method, we conducted a series of experiments. We quantify the degree of optimization for each component using improvement rates, as shown in Table 5. These quantitative measures provide a clear indication of how each module contributes to the overall performance enhancement. RAG significantly boosts prediction accuracy, highlighting the value of incorporating related data to handle knowledge updates and avoid catastrophic forgetting caused by training. Contextual information enhances domain-specific understanding, as seen in the Rouge-L results, but overly long contexts may reduce model performance. Data compression helps filter irrelevant content, ensuring more focused input and better results. This reflects the necessity of each module in improving prediction accuracy and recommendation quality for flood risk warnings. The design of the prompts used for REMFLOW is detailed in Appendix 1.2. The best results in all tables are highlighted in bold.

5 Analysis

Known data size of knowledge base To investigate the effect of known data size on performance, we present the scores under different known data sizes in Fig. 5. The performance of REMFLOW is significantly influenced by the size of the known dataset. We randomly sampled known data from the full corpus, setting various data sizes as $N = \{0, 100, 200, \ldots, ALL\}$, and conducted tests on REMFLOW based on Qwen2.5-14B-Instruct. As the size of the known data increases, the performance of REMFLOW gradually improves. Notably, using only 500 known data samples results in significant performance gains, which can be attributed to knowledge transfer facilitated by multi-factor data retrieval. When the size of known data exceeds 500 samples, the performance on both the Regional-Specific and Airport-Specific datasets stabilizes, with growth becoming more incremental, yet the overall trend remains positive.

Quality analysis of dataset We extracted structured data using rule-based matching to ensure its validity. To ensure the quality of the regional-specific factors and airport-specific factors datasets, we employ three experts to evaluate the quality of the datasets and calculate the error rate. The datasets' distribution and manual correction status are illustrated in Table 6. Our analysis reveals that high-risk data accounts for **70%** of the overall dataset. Additionally, we observe that an accuracy rate exceeding 90% in Table 3 can predominantly encompass high-risk outcomes, underscoring the effectiveness of our model in handling critical data scenarios. Manual revision includes corrections for accuracy and reasonableness, inspecting the result content to ensure it meets the requirements of the instructions, is based on prior knowledge from the input, and involves removing redundant information and hallucinations, as well as adding some undetected information. We can that the total error rate of regional-specific factors and airport-specific factors datasets is 14.8% generated by GPT-4o and then the experts fix these errors to ensure the correctness of the dataset.

Semantic similarity analysis Using the multilingual sentence-transformers[11] [30] model (SBERT), the text can be mapped into a 384-dimensional dense vector space, allowing for the measurement of semantic similarity. Given the divergent nature of large language model outputs, relying solely on common subsequences may not provide a comprehensive evaluation. Therefore, we use semantic similarity to compare the models' risk mitigation recommendations with the ground truth, providing a more robust measure of the models' generation capabilities. As shown in Fig. 6, the comparison of different models' risk mitigation recommendation generation capabilities across various regions is presented. The results indicate that the relative outcomes of semantic similarity and Rouge-L scores are largely consistent, with larger scale models demonstrating better generation capabilities, which aligns with our expectations. The detailed evaluation results are presented in Table 7.

6 Related work

Rainfall flooding and sponge airports Due to climate change, rainfall-induced flooding has become a critical issue, threatening urban infrastructure and human safety. The sponge city concept, particularly for airports, offers innovative solutions to optimize drainage systems [33, 23]. By incorporating permeable pavements, bio-swales, and retention ponds, sponge airports aim to enhance water retention and reduce flood risks. Studies have shown that these measures can effectively lower peak runoff volumes and delay flooding [26, 29]. The storm water management model (SWMM) is a widely used hydrological-hydraulic simulation

[11] https://huggingface.co/sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2.

Table 3 Evaluation results for flood risk warning generation tasks, including Accuracy, Macro F1 and Rouge-L scores across regional and airport-specific datasets

Model	Base model	Model size	Regional-specific (Acc)	Airport-specific (Acc)	Regional-specific (F1)	Airport-specific (F1)	Regional-specific (Rouge-L)	Airport-specific (Rouge-L)
Random Forest [31]	–	–	61.2	57.8	58.5	53.9	–	–
XGboost [5]	–	–	65.4	59.1	60.8	57.2	–	–
K-Nearest Neighbors [28]	–	–	55.8	51.0	47.2	44.9	–	–
LightGBM [17]	–	–	74.3	72.4	71.8	70.6	–	–
Weighted Ensemble [42]	–	–	75.0	72.1	70.3	71.5	–	–
REMFLOW + Llama	Llama3.1	8B	93.6	96.1	86.3	90.4	52.1	54.3
REMFLOW + Mistral	Mistralv0.2	7B	90.2	91.6	83.9	88.5	49.4	51.2
REMFLOW + InternLM	InternLM2	7B	88.4	87.9	81.3	85.7	47.1	48.6
REMFLOW + Qwen	Qwen2.5	7B	92.3	90.7	84.6	88.5	53.7	50.3
REMFLOW + Deepseek	Deepseekv1	7B	91.5	92.1	84.0	89.0	50.0	52.0
REMFLOW + Llama	Llama2	7B	92.0	90.5	84.3	88.0	51.0	51.5
REMFLOW + Llama	Llama2	13B	94.5	95.5	88.0	92.5	56.0	58.0
REMFLOW + Qwen	Qwen2.5	14B	**95.4**	**96.2**	**89.5**	**93.1**	**58.9**	**60.3**

All scores are reported as percentages (%)

Table 4 Ablation study results for flood risk warning generation tasks

Model	Base model	Regional-specific (Acc)	Airport-specific (Acc)	Regional-specific (F1)	Airport-specific (F1)	Regional-specific (Rouge-L)	Airport-specific (Rouge-L)
REMFLOW	Qwen2.5-14B	**95.4**	**96.2**	**89.5**	**93.1**	**58.9**	**60.3**
REMFLOW (w/o RAG)	Qwen2.5-14B	82.1	82.8	77.2	79.5	46.4	47.8
REMFLOW (w/o Context)	Qwen2.5-14B	89.1	89.9	83.8	85.5	50.6	52.2
REMFLOW (w/o Compression)	Qwen2.5-14B	92.2	92.9	86.4	88.1	53.1	54.5
Qwen2.5-Instruct	Qwen2.5-14B	80.5	81.2	75.6	77.4	45.2	46.3
REMFLOW	Llama3.1-8B	**93.6**	**96.1**	**86.3**	**90.4**	**52.1**	**54.3**
REMFLOW (w/o RAG)	Llama3.1-8B	80.6	83.1	74.0	76.8	39.6	41.8
REMFLOW (w/o Context)	Llama3.1-8B	87.3	89.8	80.3	84.1	43.8	46.2
REMFLOW (w/o Compression)	Llama3.1-8B	90.4	92.8	83.2	85.4	46.3	48.5
Llama3.1-8B-Instruct	Llama3.1-8B	79.6	81.1	72.4	74.7	38.4	40.3

The evaluated ablations include w/o (without) retrieval-augmented generation, w/o contextual information, and w/o data compression, as well as the only use of Qwen and Llama. All scores are reported as percentages (%)

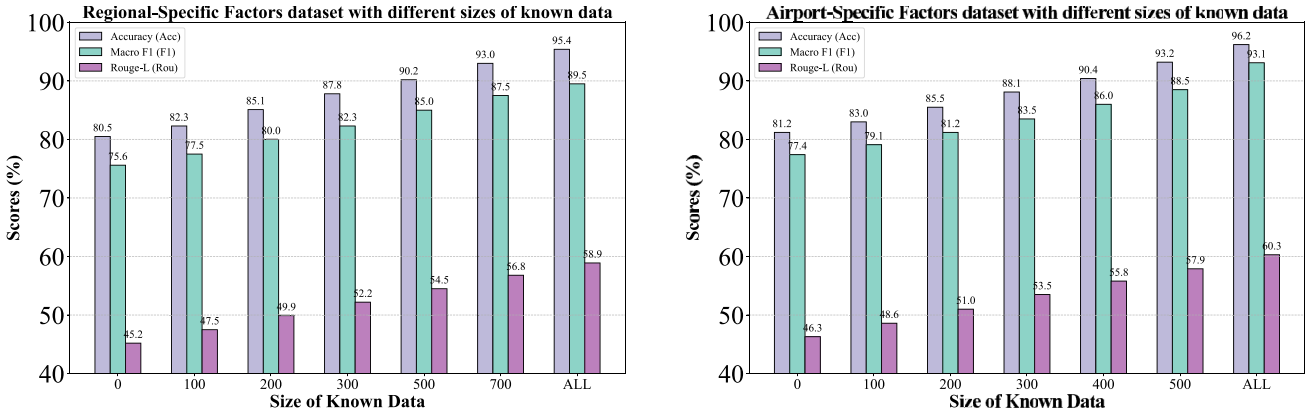
tool for modeling water flow and pollutant transport in various airport areas during rainfall events [2, 24]. Researchers use SWMM to evaluate the optimal design of permeable and impermeable surfaces to improve drainage and mitigate flood risks [22, 27]. These applications demonstrate that adjusting the layout and materials of airport surfaces can significantly enhance drainage efficiency and reduce flood risks during extreme weather events [35]. These studies provide scientific support for combining SWMM modeling with sponge airport design to develop more resilient infrastructure in the face of increasing climate uncertainties.

RAG-enhanced large language model Large language models (LLMs), such as OpenAI's Generative Pre-trained Transformer (GPT) series [1], excel in natural language processing but heavily rely on extensive datasets and complex infrastructures [40]. This dependence limits their adaptability to specialized domains, including niche scientific research, specialized engineering tasks, and sensitive data analysis. Fine-tuning smaller-scale LLMs can effectively address domain-specific long-tail challenges [16, 32], such as complex reasoning [3], security events analysis [15], IT operation [10], achieving performance comparable to larger models. Moreover, these fine-tuned models can seamlessly integrate multiple modalities, including texts [12, 37], images [36], and even videos [11], enabling more comprehensive and robust solutions across diverse applications. However, fine-tuning requires substantial computational resources and hardware, making it less efficient for domains requiring frequent updates. As a

Table 5 Performance metrics of different methods in result optimization

Method	Regional-specific (Acc)	Airport-specific (Acc)	Regional-specific (F1)	Airport-specific (F1)	Regional-specific (Rouge-L)	Airport-specific (Rouge-L)	Avg.
REMFLOW (Qwen)	18.5	18.5	18.4	20.3	30.4	30.2	19.2
RAG (Qwen)	2.0	2.0	2.1	2.7	2.7	3.2	2.5
Context (Qwen)	10.7	10.7	10.8	10.5	12.0	12.8	11.1
Compression (Qwen)	14.5	14.4	14.3	13.8	17.5	17.7	15.2
REMFLOW (Llama)	17.6	18.5	19.2	21.0	35.7	34.7	21.1
RAG (Llama)	1.3	2.5	2.2	2.8	3.1	3.7	2.6
Context (Llama)	9.7	10.7	10.9	12.6	14.1	14.6	11.9
Compression (Llama)	13.6	14.4	14.9	14.3	20.6	20.3	16.2
REMFLOW (Avg.)	**18.1**	**18.5**	**18.8**	**20.7**	**33.0**	**32.5**	**20.2**
RAG (Avg.)	1.7	2.3	2.2	2.8	2.9	3.5	2.6
Context (Avg.)	10.2	10.7	10.9	11.6	13.1	13.7	11.5
Compression (Avg.)	14.1	14.4	14.6	14.1	19.1	19.0	15.7

All values are reported as percentages (%)



(a) Regional-Specific dataset

(b) Airport-Specific dataset

Fig. 5 Evaluation results of different known data sizes

result, Retrieval-Augmented Generation (RAG) has emerged as an important research direction for knowledge enhancement [19]. RAG combines embedding models with large language models, enabling real-time retrieval and integration of relevant information from external knowledge bases, such as those used in medical diagnosis [41] and regulatory databases [6], thereby providing more accurate context in specialized domains. This approach not only reduces the cost of knowledge updates but also enhances the model's generalization capabilities, leading to superior performance on domain-specific tasks [14]. The use of RAG demonstrates that, in handling vertical domain challenges, it can efficiently utilize hardware resources while achieving performance comparable to or even exceeding that of larger models. RAG technology is also an effective solution for mitigating hallucinations in large language models [25].

7 Conclusion

In this work, we present REMFLOW, a novel flood early warning system tailored for sponge airports, integrating the SWMM model with multi-factor data analysis. By combining rainfall, runoff, and environmental parameters, our method enhances the accuracy of flood prediction and

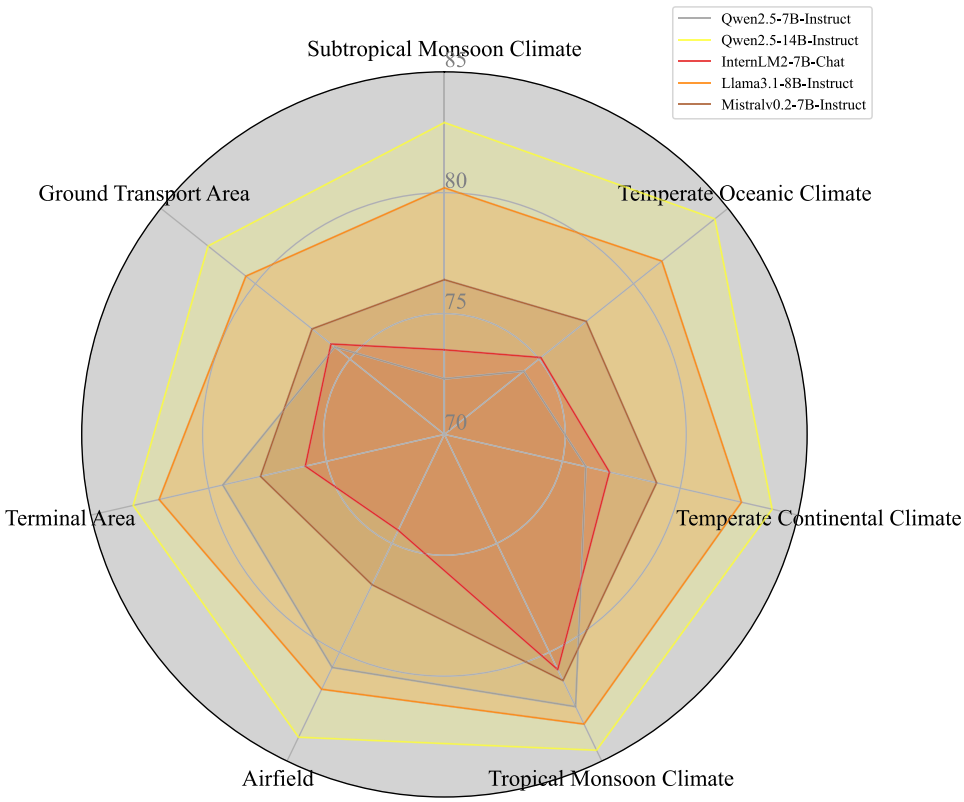


Fig. 6 Comparison of different models' risk mitigation recommendation generation capabilities across various regions

Table 6 Distribution and correction of regional-specific factors and airport-specific factors datasets

Region category	|High|	|Moderate|	|Low|	|High (C)|	|Moderate (C)|	|Low (C)|	Ratio (%)
Regional-specific factors datasets							
Subtropical monsoon climate	173	2	0	17	0	0	9.7
Temperate oceanic climate	413	2	149	41	1	15	10.1
Temperate continental climate	242	1	8	48	0	2	19.9
Tropical monsoon climate	1	9	0	0	0	0	0
Airport-specific factors datasets							
Airfield	74	124	2	14	24	1	19.5
Terminal area	170	10	20	33	3	4	20.0
Ground transport Area	182	12	6	32	2	0	17.0
Total	1255	160	185	185	30	22	14.8

|High| denotes the number of High-risk alert samples in datasets while |High(C)| denotes the number of samples need to be corrected

management. Leveraging large language models (LLMs) within a retrieval-augmented generation (RAG) framework, REMFLOW effectively incorporates unstructured environmental data and contributes to the development of two multi-factor structured datasets, regional-specific factors and airport-specific factors. At the same time, it improves the methods of knowledge update and enhancement in the application of LLMs in vertical domains, offering insights for

Table 7 Semantic similarity scores (%) across different models and regions

Region	Qwen2.5-7B	Qwen2.5-14B	InternLM2-7B	Llama3.1-8B	Mistralv0.2-7B	Deepseekv1-7B	Llama2-7B	Llama2-13B
Subtropical monsoon climate	72.3	**82.9**	73.5	80.2	76.4	78.2	80.8	81.5
Temperate oceanic climate	74.2	**84.3**	75.1	81.5	77.5	79.8	80.5	83.4
Temperate continental climate	76.0	**85.0**	77.0	83.6	79.0	81.8	82.1	83.6
Tropical monsoon climate	78.5	**84.5**	76.8	82.3	78.3	80.9	81.0	82.4
Airfield	80.7	**83.9**	74.4	81.7	76.9	80.0	82.7	81.7
Terminal area	79.4	**83.2**	75.9	82.1	77.8	80.1	81.7	82.2
Ground transport area	75.8	**82.5**	76.0	80.5	77.0	79.1	80.3	81.5

the application of LLMs in flood early warning for sponge airports. Comparative experiments demonstrate that REM-FLOW surpasses traditional machine learning approaches, offering a more reliable and intelligent solution for flood forecasting in sponge airport environments.

Appendix 1: Instructions

1.1: Date generation prompt

The following prompts are designed for the generation of regional-specific factors data.

[Instruction]
You are an expert in sponge airport rain and flood warning data analysis. There is a dataset that requires analysis and generation based on data from various climate regions. I will provide some descriptions of the rainfall and flood conditions, and the dataset content should align with the reference sample provided in the "Notice". Using the information described in the "Related Documents" and "Related Data", selectively apply the relevant descriptions and your professional knowledge to provide realistic and reliable data in JSON format. Ensure that the generated data is as accurate as possible, and keep the units consistent without including them explicitly, as default units are assumed.
[Related Documents]
Precipitation
• Definition: Refers to the total amount of rainfall in a specific area, essential for assessing flood risk and the effectiveness of water management systems.
• Measurement: Typically measured in millimeters (mm).

Date
• Definition: The specific date when the weather data is recorded. This helps in correlating weather patterns and assessing potential risks.
• Measurement: Expressed as a calendar date (e.g., YYYY-MM-DD).

Regional Climate Type
• Definition: The general classification of the climate in the region, such as "Subtropical Monsoon Climate." It helps in predicting regional weather patterns and responses to weather events.
• Measurement: Described using standard climate classifications (e.g., subtropical, temperate).

Historical Weather Conditions
• Definition: Information about past weather patterns, including rainfall frequency and intensity. Used to identify trends and predict future events.
• Measurement: Quantified using historical data on rainfall events, often summarized by averages or extremes (e.g., maximum rainfall in the past 5 years).

Permeable Pavement Area
• Definition: The surface area covered by permeable materials that allow water to pass through, reducing runoff and aiding infiltration in flood management.
• Measurement: Measured in square meters (m^2).

Rainwater Harvesting System Capacity
• Definition: The maximum volume of rainwater that can be collected and stored by the system. It indicates the system's ability to manage excess rainfall.
• Measurement: Measured in liters (L).

Vegetation Coverage Rate
• Definition: The percentage of the area covered by vegetation. Vegetation helps absorb water, reducing runoff and mitigating flood risks.
• Measurement: Expressed as a percentage (%) of the total area.

The following prompts are designed for the generation of airport-specific factors data.

Retention Pond Capacity
• Definition: The maximum volume of water that can be held by retention ponds. These ponds are used to temporarily store stormwater runoff, reducing the risk of flooding.
• Measurement: Measured in liters (L).

Groundwater Recharge Rate
• Definition: The volume of water added to the groundwater system on a monthly basis. This is critical for maintaining groundwater levels, especially during dry periods.
• Measurement: Measured in cubic meters per month (m^3/month).

[Related Data]
{Replace with related data here}
[Notice]
The generated results are expressed in the form of a json format as the example.
{Sample data}
[Output]

1.2: Flood warning prompt

The following prompts are designed for the early warning system based on LLM.

[Instruction]
You are an expert in sponge airport rain and flood warning data analysis. There is a dataset that requires analysis and generation based on data from different areas of the same sponge airport. Now you need to generate **Airfield/Terminal Area/Ground Transport Area** data. I will provide some descriptions of the rainfall and flood conditions, and the dataset content should align with the reference sample provided in the "Notice". Using the information described in the "Related Documents" and "Related Data", selectively apply the relevant descriptions and your professional knowledge to provide realistic and reliable data in JSON format. Ensure that the generated data is as accurate as possible, and keep the units consistent without including them explicitly, as default units are assumed.
[Related Documents]
Storm Event Simulation
• Definition: Simulates storm events using the SWMM model to analyze rainfall distribution and water accumulation in different airport areas.
• Measurement: Based on historical rainfall data and current forecasts, typically expressed in millimeters per hour (mm/h).

Flooding and Waterlogging Analysis
• Definition: Uses the SWMM model to simulate water accumulation, analyzing depth and duration of flooding in various airport zones.
• Measurement: Water depth in centimeters (cm) or meters (m), duration in hours (h).

Drainage System Process
• Definition: Analyzes and monitors the efficiency of the airport's drainage system during storms, including pipe flow rate and drainage speed.
• Measurement: Pipe flow rate in liters per second (L/s) or cubic meters per second (m³/s), drainage speed in meters per second (m/s).

The maximum volume of rainwater that can be collected, measured in liters. It indicates the system's capacity to manage excess rainwater.

Vegetation Coverage Rate
The percentage of the area covered by vegetation. Vegetation absorbs water, reduces runoff, and mitigates flood risks.

Retention Pond Capacity
he maximum water volume retention ponds can hold, measured in liters. They manage stormwater by temporarily storing runoff.

Groundwater Recharge Rate
The volume of water added to the groundwater system per month, measured in cubic meters. It is crucial for maintaining groundwater levels, especially in dry periods.

[Reference data]
{Replace with Retrieval data here}
[Notice]
The generated results are expressed in the form of a json format as the example.
{Sample data}
[Output]

Appendix 2: Datasets

2.1: Dataset sample

This section presents examples of regional-specific factors and airport-specific factors respectively.

```
# Regional-Specific Factors

{
  "input": {
    "Precipitation": "50",
    "Date": "2024-07-28",
    "Regional Climate Type": "Subtropical Monsoon Climate",
    "Historical Weather Conditions": "This region frequently
       experiences short-duration heavy rainfall during summer; the
       maximum precipitation in the past five years was 100",
    "Permeable Pavement Area": "40,000",
    "Rainwater Harvesting System Capacity": "500,000",
    "Vegetation Coverage Rate": "60",
    "Retention Pond Capacity": "200,000",
    "Groundwater Recharge Rate": "10,000"
  },
  "output": {
    "Current Risk Level": "Moderate",
    "Recommended Actions": [
      "Monitor the water storage level of the rainwater harvesting
         system and increase storage when necessary",
      "Open some sections of the retention pond to handle potential
         ongoing rainfall",
      "Given the current moderate risk level, it is recommended not
         to adjust flight schedules yet, but closely monitor weather
         changes and prepare contingency plans for flight adjustments
          if needed"
    ]
  }
}
```

```
# Airport-Specific Factors

## Airfield

{
  "input": {
    "area": "Airfield",
    "storm_event_simulation": {
      "rainfall_intensity": "85",
      "simulation_duration": "2",
      "maximum_water_accumulation": "20"
    },
    "flooding_and_waterlogging_analysis": {
      "water_depth": "15",
      "water_accumulation_duration": "5"
    },
    "drainage_system_process": {
      "main_drainage_pipe_flow_rate": "3.0",
      "drainage_speed": "1.5"
    },
    "real_time_hydrological_monitoring": {
      "real_time_rainfall": "80",
      "water_level": "1.0",
      "water_quality": {
        "pH": "7.0",
        "dissolved_oxygen": "6.8"
      }
    },
    "waterlogging_point_warning_indicators": {
      "warning_threshold": "15"
    }
  },
  "output": {
    "current_risk_level": "High",
    "recommended_actions": [
      "Immediately initiate enhanced drainage protocols to handle the
          high water accumulation, focusing on areas with 15 cm water
          depth or more.",
      "Monitor the drainage system closely to ensure it is operating
         at maximum capacity, considering additional pumps or
         emergency drainage routes if necessary.",
      "Prepare to delay or reroute flights to prevent potential
         safety hazards due to water accumulation on runways.",
      "Continuously update real-time hydrological data and adjust
         emergency response plans based on current water levels and
         rain intensity."
    ]
  }
}
```

```
# Airport-Specific Factors

## Terminal Area

{
    "input": {
            "area": "Terminal Area",
            "storm_event_simulation": {
                "rainfall_intensity": "70",
                "simulation_duration": "3",
                "maximum_water_accumulation": "10"
            },
            "flooding_and_waterlogging_analysis": {
                "water_depth": "8",
                "water_accumulation_duration": "4"
            },
            "drainage_system_process": {
                "main_drainage_pipe_flow_rate": "2.0",
                "drainage_speed": "1.0"
            },
            "real_time_hydrological_monitoring": {
                "real_time_rainfall": "65",
                "water_level": "0.5",
                "water_quality": {
                    "pH": "7.3",
                    "dissolved_oxygen": "6.7"
                }
            },
            "waterlogging_point_warning_indicators": {
                "warning_threshold": "8"
            }
        },
        "output": {
            "current_risk_level": "Moderate",
            "recommended_actions": [
                "Continue monitoring the drainage system and prepare
                   to activate additional drainage pumps if water
                   levels approach the 8 cm threshold.",
                "Ensure clear communication with terminal operations
                   to avoid disruption due to minor flooding, and
                   prepare to relocate passengers or services if
                   necessary.",
                "Monitor real-time water quality and adjust the
                   drainage management plan to prevent waterlogging
                   around sensitive infrastructure.",
                "Maintain regular updates with the weather forecast
                   team to anticipate any changes in storm patterns."
            ]
        }
}
```

```
# Airport-Specific Factors

## Ground Transport Area

{
        "input": {
            "area": "Ground Transport Area",
            "storm_event_simulation": {
                "rainfall_intensity": "75",
                "simulation_duration": "2.5",
                "maximum_water_accumulation": "12"
            },
            "flooding_and_waterlogging_analysis": {
                "water_depth": "10",
                "water_accumulation_duration": "3"
            },
            "drainage_system_process": {
                "main_drainage_pipe_flow_rate": "2.8",
                "drainage_speed": "1.3"
            },
            "real_time_hydrological_monitoring": {
                "real_time_rainfall": "70",
                "water_level": "0.7",
                "water_quality": {
                    "pH": "7.1",
                    "dissolved_oxygen": "6.6"
                }
            },
            "waterlogging_point_warning_indicators": {
                "warning_threshold": "10"
            }
        },
        "output": {
            "current_risk_level": "Moderate",
            "recommended_actions": [
                "Implement drainage measures to prevent water depth
                   from exceeding 10 cm, focusing on critical
                   transport routes.",
                "Monitor the effectiveness of the drainage system and
                    be prepared to deploy additional resources if
                   water levels rise.",
                "Inform ground transport services of potential delays
                    or rerouting due to water accumulation.",
                "Maintain regular communication with airport
                   operations and emergency services to ensure quick
                   response to any escalations."
            ]
        }
}
```

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Author Contributions G. W. led the primary writing of the manuscript, organizing the content and performing data analysis. Y.L. provided guidance throughout the research process, offering critical corrections and revisions to the manuscript. S.L. and L.Z contributed to the research by providing essential data support. L.Y. secured funding for the project and facilitated necessary resources for the study. All authors reviewed and approved the final version of the manuscript.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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